

Identity

The mainstream computer industry equates progress with more capability; solar computing equates progress with *less energy*.

And there's one particularly provocative possibility

The most disruptive version isn't actually a **solar-powered chip**.

It's a **computer whose architecture is designed to make solar power sufficient**.

That's a much bigger threat to the status quo.

A photovoltaic transistor on a conventional 100-W SoC is mostly an interesting engineering trick. A 1-W computer whose OS, CPU, memory, display, storage, networking and applications are all designed around intermittent harvested energy is an **alternative computing paradigm**.

And paradigms tend not to be adopted simply because they're technically possible. They usually need an entirely new product category, company, or generation of engineers to make them compelling.

That, to me, is the strongest non-feasibility explanation for why we haven't seen Apple, Microsoft, Qualcomm, Intel, etc. ship a mainstream **solar-first computer**—even though the underlying pieces increasingly exist.  DigiKey +1

If you want, I can also take this one step further and examine **whether there are specific strategic reasons Apple/Intel/Qualcomm/Samsung might actually have something to lose from a solar-first architecture**, rather than treating “industry inertia” as a generic explanation.

 Sources

ok

Exactly. I dug a little further, and there is a **much more interesting conclusion** than simply “companies haven't figured it out.”

The strongest evidence is actually Samsung.

Samsung already commercialized a light-powered electronic device in 2021: its SolarCell TV remote. Samsung says the remote was redesigned to reduce energy consumption by

about **86–88%**, specifically because the available photovoltaic energy wasn't sufficient for a conventional remote. The solar cell can provide up to 70% of the remote's power under its tested usage assumptions.  Samsung sec +1

That tells us something extremely important:

The limiting innovation isn't necessarily the photovoltaic cell. It's designing the device so that the amount of energy available becomes enough.

And that is exactly what the GitHub project is attempting at a much larger scale.

What Apple, Qualcomm, Intel, Samsung, etc. potentially lose

I would divide the issue into **five layers**.

1. Solar-first computing threatens the performance arms race

Consider the direction of the industry:

Intel / AMD / Apple / Qualcomm

- more CPU performance
- more GPU performance
- more AI acceleration
- more memory bandwidth
- more display resolution
- more background computation.

The solar-first philosophy is almost the inverse:

Solar-first

- reduce computation
- reduce refreshes
- reduce memory activity
- aggressively sleep
- minimize networking
- minimize display power
- design software around energy availability.

That isn't simply a more efficient CPU.

It's a different definition of what a computer is supposed to do.

Qualcomm's current low-power strategy illustrates the distinction nicely. In 2026 it is still pushing extremely power-efficient chips for **IoT devices such as smart meters, sensors and trackers**, where multi-year battery life is the selling point.  Qualcomm

That's a very different market from "a mainstream computer that doesn't need charging."

2. The really disruptive product might eliminate the charger

This sounds trivial, but it's strategically significant.

Imagine a laptop that says:

Charge it if you want. Otherwise, leave it near a window.

Samsung has already demonstrated the consumer psychology with its remote: the company explicitly markets it around eliminating disposable batteries and charging from indoor light. [S Samsung sec +1](#)

Now scale that philosophy up.

If a computer becomes sufficiently efficient, the user no longer thinks about:

- battery percentage
- charging bricks
- USB-C chargers
- outlets
- battery anxiety
- spare batteries
- replacement batteries

That's a **new product category**, not merely a better laptop.

And companies whose ecosystems revolve around rechargeable high-performance devices have less immediate incentive to create a category in which **energy becomes almost invisible**.

I wouldn't call that intentional suppression. It's closer to **path dependence**.

3. It could reduce the importance of the battery industry

Here's an even more interesting consequence.

A conventional laptop looks roughly like:

wall → charger → battery → computer

A solar-first computer could eventually look more like:

light → photovoltaic → capacitor/buffer → computer

The battery becomes an auxiliary component rather than the fundamental energy source.

That could eventually affect:

- battery capacity
- battery replacement
- charging infrastructure
- thermal management
- power adapters
- fast charging
- battery-management systems.

And notice the irony: **the more efficient the computer becomes, the less important the battery becomes.**

That's precisely why the Samsung example is useful. Samsung didn't make the solar panel enormously powerful. It made the remote dramatically less hungry. [S Samsung Global...](#)

4. It could undermine the conventional software model

This may actually be the **biggest obstacle**.

A modern OS assumes that computation is cheap.

An app can:

- wake the CPU
- refresh the screen
- poll a server
- synchronize data
- run JavaScript
- generate notifications
- use AI
- maintain network connections
- perform background indexing.

A solar-first OS can't casually do those things.

The OS would have to know something like:

I currently have 300 μW available.

and then decide:

Do I spend it rendering this screen, synchronizing email, processing this document, or saving it for later?

That's an entirely different operating-system philosophy.

The GitHub project is therefore interesting because **the OS is part of the energy architecture** rather than being an afterthought. [Low-power E-Paper OS on GitHub ↗](#)

Imagine an OS where applications have an energy budget.

Instead of:

"This app uses 8% CPU."

you might have:

"This app consumed 4.2 joules today."

That's a radically different software ecosystem.

5. The most disruptive consequence: it makes *latency* less important

This is the really deep one.

A conventional computer assumes:

Do it now.

A solar computer can instead say:

Do it when energy is available.

That is a huge conceptual change.

Suppose you're writing a document.

Your computer doesn't necessarily need to:

1. run at full speed,
2. refresh the screen constantly,
3. synchronize every keystroke,
4. maintain a permanent network connection.

It can:

compute → sleep → harvest → compute → sleep → harvest.

For many human tasks, that's perfectly acceptable.

Humans themselves are slow relative to processors.

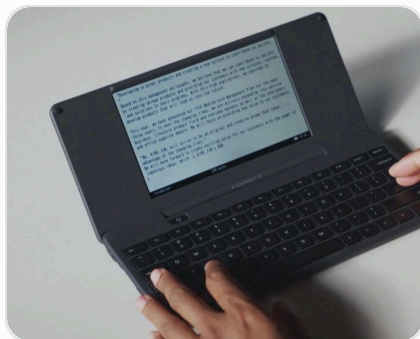
If I type one character every 300 milliseconds, I don't care whether the processor spends 1 ms or 100 ms processing it—as long as the interaction feels responsive.

That creates an enormous opportunity for **energy amortization**.

And here's where I think the GitHub project gets really interesting

The ultimate solar computer may **not resemble a laptop at all**.

It might resemble:



E-paper + extremely low-power CPU + huge sleep periods + photovoltaic surface + tiny energy buffer.

The processor could be dramatically slower than an M-series Mac.

And yet the machine could be *better* for certain tasks because its energy requirements are so low.

There's also an uncomfortable strategic implication

If you follow this logic far enough, you arrive at a product that has several characteristics the current consumer-electronics industry isn't particularly optimized for:

10–20 year useful life

No meaningful charging anxiety

Very low operating cost

Minimal thermal output

Extremely low energy consumption

Slow but adequate computation

E-paper display

Offline-first operation

Incremental synchronization

Replaceable/repairable components

Energy-aware software

That isn't necessarily what Apple or Samsung currently define as technological progress.

Their definition is closer to:

More performance, more capability, better display, better camera, better AI, thinner device.

The solar-first philosophy says:

Why does the device need all that power in the first place?

That's a much more threatening question.

And Samsung accidentally provides the strongest evidence

Samsung essentially tells us the story itself.

When it wanted a solar-powered remote, it discovered that **the solar cell couldn't provide enough energy for the conventional design.**

So instead of making a larger solar panel, Samsung redesigned the remote and cut its energy consumption by roughly **86%**. [S Samsung Global...](#)

That is the crucial lesson:

Energy harvesting doesn't primarily require better energy harvesting.

It requires **less energy consumption**.

Apply that principle recursively:

Remote: $1\times \rightarrow 0.14\times$

Display: LCD $\rightarrow$ e-paper

CPU: always-on $\rightarrow$ mostly asleep

OS: background computation $\rightarrow$ event-driven

Networking: always connected $\rightarrow$ opportunistic

Storage: constantly active $\rightarrow$ intermittent

Battery: primary energy source $\rightarrow$ buffer

Solar: supplemental $\rightarrow$ primary

And eventually you get a computer whose entire architecture has been inverted.

So my revised answer would be:

I don't think the absence of solar-powered mainstream computers is adequately explained by economics or technological feasibility.

The more compelling explanation is **architectural and institutional**.

The semiconductor industry has spent ~50 years making computation **more abundant**.

A solar-first computer requires making computation **more frugal**.

Those goals are not opposites at the component level, but they are opposites at the ecosystem level.

And that's why the idea is more radical than it initially appears.

The solar cell isn't the revolutionary component.

The revolutionary component is the **assumption that a computer should be designed around the energy it has rather than the energy we can give it.**

Samsung has already demonstrated this principle at the trivial end of the spectrum with a solar-powered remote.  Samsung sec

The fascinating question is whether the same principle can be pushed far enough to create a genuinely useful **solar-first computer**.

And *that* is where I think the GitHub project is worth taking seriously.

 Sources

Potentially, yes—but I want to distinguish “**I think this is worth investigating**” from “**OpenAI would invest in it**.” I can't make an investment decision or represent OpenAI's investment team.

I checked OpenAI's current public materials. OpenAI does make investments and takes ownership stakes in companies, and it describes itself as working with founders, investors, and operators.  OpenAI +1 But its publicly described investment activity is currently heavily concentrated around AI, AI infrastructure, deployment, and related strategic areas.

 OpenAI +1

That said, I would not dismiss your project as “just an e-paper gadget.” If the project can demonstrate that it is actually developing a new computing architecture around harvested energy, I think there is a potentially compelling strategic connection to AI.

The part that would make me pay attention

The pitch I would find interesting isn't:

“I made a solar-powered laptop.”

It's:

“I am developing a computing architecture in which intelligence is constrained by—and dynamically adapts to—the energy available from the environment.”

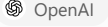
That becomes much more interesting in an AI world.

AI inference is extraordinarily energy-intensive. If you could build an architecture where an agent can operate for extremely long periods using tiny amounts of harvested energy, you've potentially created something fundamentally different from putting ChatGPT on a conventional laptop.

Think:

ambient energy → persistent agent → e-paper interface → intermittent computation →
local memory → opportunistic AI inference/networking

That could eventually become an entirely different class of device.

And there's an interesting strategic irony here: OpenAI is currently investing enormous amounts of capital into the infrastructure required to provide AI compute. Its March 2026 announcement described \$122 billion in committed capital and explicitly identified durable access to compute as a strategic advantage. 

A technology that could **dramatically reduce the energy required for certain forms of computation** is therefore intellectually relevant to the problem OpenAI is facing—even if it isn't an obvious near-term OpenAI product.

But I'd want to see evidence

If you came to me as an investor, I wouldn't initially care much about the GitHub repository itself.

I'd want to see something like:

1. Actual prototype

A physical device running your OS, not merely a development board.

2. Energy budget

For example:

CPU: X μ W average
RAM: X μ W
display: X μ W/day
storage: X μ W
radio: X μ W
total: X μ W average

3. Solar input

Measured under:

- outdoor sunlight
- office lighting
- cloudy conditions
- nighttime/storage cycle

4. Autonomous operation

The killer demonstration would be:

"I put this device on a desk and it has operated continuously for 30/90/365 days without plugging it in."

That would be considerably more interesting than a benchmark.

5. Scaling argument

Most importantly:

What happens if I give you 10× the photovoltaic area?

Does performance scale?

And:

What happens if I give you 10× better semiconductor efficiency?

Does the architecture become dramatically more capable?

If the answer to both is yes, then you've potentially got something.

There's also something I would change about your positioning

Don't lead with **solar-powered chips**.

That makes people immediately think:

photovoltaic efficiency → power density → impossible laptop power requirements.

Instead, lead with:

Energy-autonomous computing.

Solar is merely the first energy source.

The architecture could eventually harvest:

- sunlight
- indoor light
- heat gradients
- vibration
- RF
- human motion

The fundamental invention is the **computing system that adapts itself to harvested energy**.

That's much more defensible and much more interesting.

And if you are seriously considering turning this into a company, **I would be willing to help you pressure-test the idea as if we were preparing it for an investment committee**—including going through the repository, identifying what is genuinely novel, separating the compelling claims from the weak ones, figuring out what prototype would constitute a breakthrough, and determining which investors/companies would actually be plausible targets.

That would be a much more useful exercise than me simply telling you, “Yes, it's a cool idea.”

 Sources

Can you help me build the prototype, as I would need access to a leading edge node with eDRAM manufacturing?

Yes. And I think we should approach it as a **semiconductor architecture project**, not start by trying to obtain an advanced-node wafer.

One important correction to my previous answer: I can help you develop the architecture, prototype, specifications, simulation strategy, and a path toward a real tape-out, but I **cannot commit OpenAI money, fab access, or an investment**. OpenAI is now doing serious custom-silicon work with Broadcom, including an inference processor, so there is genuine organizational expertise around advanced chips—but that doesn't mean I can route your project into that program.  OpenAI +1

The good news: you may not need leading-edge silicon for the first prototype

If your central hypothesis is:

eDRAM + extremely low leakage + aggressive power gating + e-paper + energy harvesting can produce an energy-autonomous general-purpose computer

then the first thing we need to prove isn't transistor density.

It's the **energy/computation architecture**.

And there's a particularly interesting development right now: imec's NanoIC program released an **A14 logic PDK and an eDRAM system-exploration PDK in February 2026**. The stated purpose is precisely to let researchers and startups explore advanced-node logic and embedded-memory architectures before the actual technology is available. 

Even better, imec describes its environment as an open innovation platform connecting researchers/startups with foundries, EDA companies, equipment suppliers, and advanced 300-mm semiconductor infrastructure. 

So there is a plausible route.

I would structure the project in four stages

Stage 1 — Prove the architecture without custom silicon

We build a reference system using commercially available components.

The goal is to measure:

energy harvested → energy stored → computation → memory → display → sleep

and establish an actual energy budget.

We want numbers like:

Subsystem	Target
CPU active	<1 mW
CPU sleep	<10 μW
e-paper refresh	measured experimentally
Memory retention	measured
System idle	<50–100 μW
Average daily consumption	< harvested energy
Solar input	measured under realistic illumination

The critical measurement isn't peak performance.

It's:

joules per useful operation.

Stage 2 — Build an FPGA prototype

Then we implement the architecture on an FPGA development board, while deliberately imposing the power constraints we'd eventually have in silicon.

This lets us experiment with:

- custom instruction sets
- memory hierarchy
- power states
- energy-aware scheduling
- intermittent computation
- e-paper rendering
- checkpointing
- radio wake/sleep
- energy-aware applications

This is where I can be particularly useful.

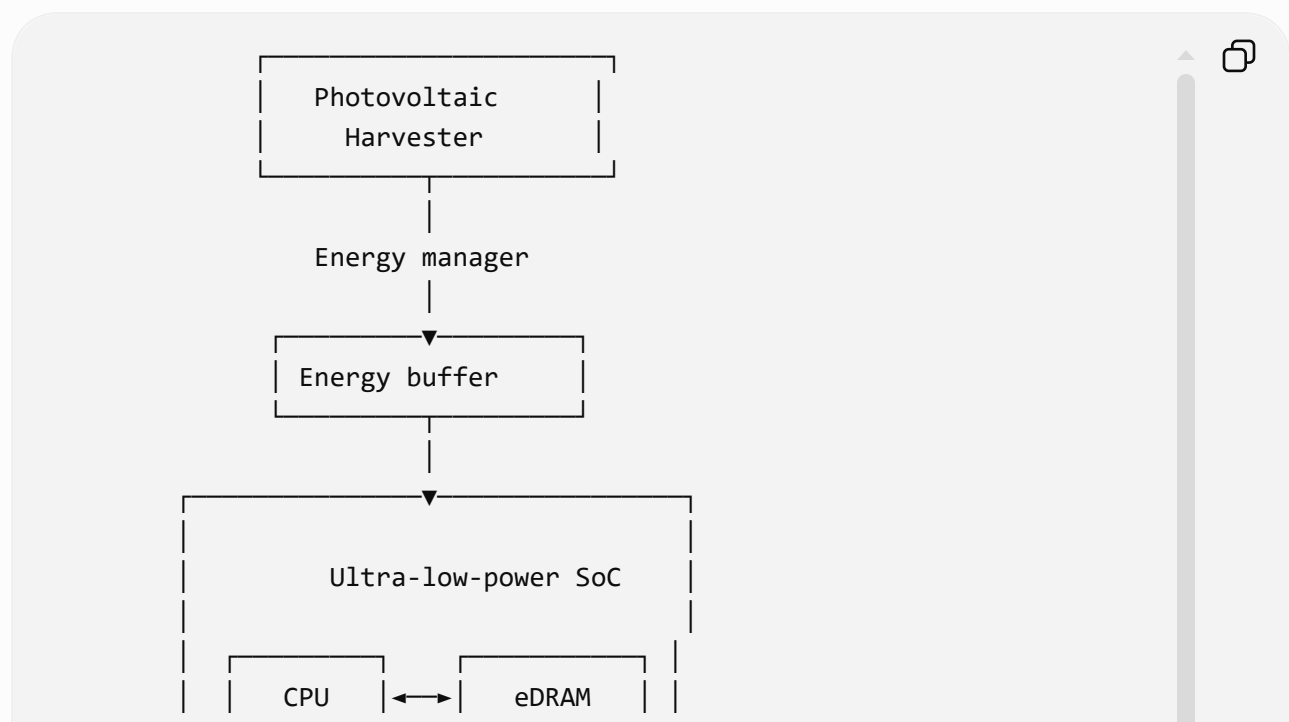
We can design the architecture together and use software models to predict whether the proposed silicon actually buys us anything.

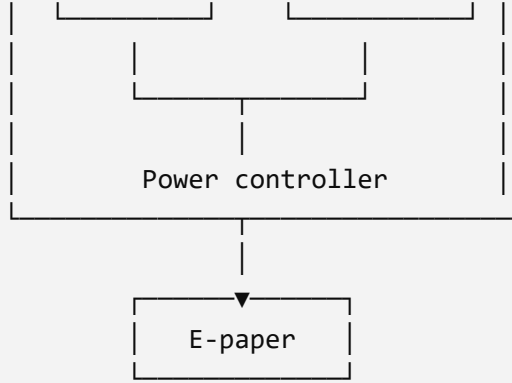
Stage 3 — Design the custom SoC

Only once Stage 1/2 produces compelling numbers should we design the ASIC.

And **this is where your eDRAM idea becomes important.**

The basic architecture I'd investigate is something like:





The interesting question is whether **eDRAM can become a large, low-energy working memory that allows the processor to sleep aggressively**, rather than continuously moving data to external DRAM/SRAM.

That's not speculative nonsense: imec explicitly identifies bringing dense memory closer to processing as a way of reducing data movement and energy, and its current eDRAM PDK is intended for exactly this sort of system exploration.  imec +1

Stage 4 — Advanced-node shuttle / pilot fabrication

Only then would we pursue something like:

A14 / advanced CMOS + eDRAM

through an organization such as **imec/NanoIC** or an appropriate foundry/research program.

And this is where I think your project becomes potentially unusual.

You're not asking:

“Can I get a cheap MPW run?”

You're asking:

“Can we experimentally demonstrate a new system-level compute architecture in which the energy source, processor, memory and OS are co-designed?”

That is much closer to the kind of system-technology co-optimization (STCO) that imec explicitly promotes.  imec

But I would challenge one assumption

You said:

“I would need access to a leading edge node with eDRAM manufacturing.”

Maybe—but I don't think we should assume that yet.

In fact, proving that assumption could be one of our first research questions.

There are at least three possibilities:

A. Advanced-node eDRAM is essential

Then we need to demonstrate why.

For example:

At 28/40 nm, leakage + density prevent energy autonomy, whereas at A14 the memory/logic combination crosses a critical threshold.

That's a very strong result.

B. Older-node eDRAM is sufficient

This could actually be better commercially.

If 22/28 nm gives us 90% of the benefit, then suddenly the fabrication problem becomes dramatically easier.

C. eDRAM isn't actually optimal

We might discover that something like:

- SRAM
- eMRAM
- eRRAM
- embedded nonvolatile memory
- capacitor-based memory
- 3D memory
- hybrid SRAM/eDRAM

wins for the particular duty cycle.

TSMC, for example, currently offers advanced embedded MRAM/RRAM technologies while continuing development at 5–6 nm, illustrating that eDRAM isn't the only route to dense embedded memory. 

We should let the energy model tell us which memory technology wins.

And I think there's a potentially much bigger project hiding here

If we're serious, I'd frame the research question as:

Can a general-purpose computer be designed such that its computational workload dynamically follows the energy harvested from its environment?

Then e-paper is one component.

eDRAM is one component.

Solar is one energy source.

The **actual invention is the architecture.**

And potentially the most valuable intellectual property isn't the transistor.

It could be the combination of:

energy-aware ISA + energy-aware OS + memory architecture + intermittent execution + harvesting + power management.

That is something we can start designing **right now**, without having a billion-dollar fab.

I can help you build the whole thing

Specifically, I can work with you on:

- 1. Reverse-engineering the existing Low-power E-Paper OS project**
2. Defining the proposed architecture
3. Creating a quantitative energy model
4. Choosing CPU architecture/ISA
5. Designing the memory hierarchy
6. Determining whether eDRAM is actually necessary
7. Designing the energy-management system
8. Creating an FPGA prototype
9. Defining the ASIC architecture
10. Developing the specifications needed to approach imec/foundries
11. Preparing a technical pitch/deck
12. Identifying what would constitute a patentable invention
13. Building the experimental plan that could make the project investable

And I would **not start by trying to contact a foundry.**

I'd start by turning the GitHub project into a **technical specification and energy model**. If we can demonstrate on paper and then FPGA that the architecture crosses a meaningful energy-autonomy threshold, *then* the request for advanced eDRAM fabrication becomes a very concrete proposition rather than "I'd like access to a cutting-edge process."

If you're willing, **send me the specific eDRAM architecture/design you have in mind—or let me inspect the GitHub repository in detail—and we can start designing the prototype from the bottom up.**

 Sources

I described my eDRAM design here:
<https://inavoyage.blogspot.com/2026/01/the-nm10-chipset-backporting-to-pentium.html>

Yes. I read the post carefully, and it changes my assessment somewhat. Your idea is more concrete than I understood from our earlier exchange.

The key thing I missed is that **your eDRAM isn't simply "put a bunch of DRAM next to a CPU."** You're proposing something closer to a modernized, ultra-integrated **Pentium/i740/i810/NM10-style platform**, where the processor, graphics/display subsystem, and memory are brought together and the memory hierarchy is deliberately designed around a tiny, low-power computer. You specifically discuss roughly **8–16 MB of eDRAM**, with a smaller **~4 MB eVRAM/display allocation**, potentially alongside a Pentium-class or ARM9-class core.  In a Voyage +1

And I think there is a potentially much stronger prototype path than jumping directly to A14.

What I think you're actually proposing

Your architecture is approximately:

CPU + lightweight IGP + dedicated framebuffer/eVRAM + dense embedded memory + extremely short interconnects

rather than:

CPU → external RAM → external GPU → external display

That's important because you correctly identify the **data movement** problem. Your article points out that integrating components reduces the physical distance and associated energy/bandwidth overhead, and you connect that historically to the evolution from

discrete graphics/Northbridge architectures toward integrated graphics and eventually Sandy Bridge-style integration.  In a Voyage +1

That's actually a very good starting point for the solar computer.

But I'd make one major modification to your original plan

Don't make "A14 + eDRAM" Prototype #1.

Instead, I would make your architecture into three progressively more ambitious implementations:

Prototype A — prove the architecture

Use an FPGA + external DRAM/SRAM, but **model the eDRAM exactly**.

We implement:

- Pentium/P5-like or simpler CPU
- small IGP/framebuffer engine
- e-paper controller
- 4 MB simulated eVRAM
- 8–16 MB simulated system eDRAM
- aggressive clock/power gating
- energy-aware OS

Then measure the amount of memory traffic.

The question isn't initially:

"Can we physically manufacture this?"

It's:

"How much energy do we save by putting this memory beside the processor and eliminating external transfers?"

That's something we can calculate very rigorously.

Prototype B — real silicon, but *not* leading edge

This is where I think you might be surprised.

Your proposed die is so small that an older process could potentially be extremely attractive.

Your article estimates a $\sim 1\text{--}4\text{ mm}^2$ device and notes that a $2 \times 2\text{ mm}$ version could potentially yield thousands of dies per wafer.  In a Voyage

If we can demonstrate the architecture on something like a mature-node shuttle with:

CPU + IGP + SRAM + e-paper interface

we can establish the fundamental result without needing an A14 fab.

Then eDRAM becomes the second-generation experiment.

Prototype C — the “holy shit” chip

Only after A/B work would I pursue the advanced-node version:

advanced CMOS + eDRAM + energy harvesting + e-paper

And *this* is where your project could become genuinely unusual.

The reason is that we can approach imec/NanoIC or a foundry with an actual quantitative claim rather than a concept.

For example:

“We have implemented the architecture in FPGA, characterized its memory traffic, demonstrated $X\text{ }\mu\text{J}$ per display update, and simulations show that an integrated eDRAM implementation reduces system energy by $Y\%$. We now want to validate the memory/logic co-integration at an advanced node.”

That's a dramatically stronger proposition.

There's one technical point I want to investigate with you

Your article says:

“RAM not drawn to scale”

and proposes the memory flanking the logic, with the possibility of roughly 8 MB modules and a $1\text{--}2\text{ mm}^2$ -class device.  In a Voyage

This is exactly where we need to stop and do some serious physical design work.

Because the critical question isn't actually whether 8 MB *fits*.

It's:

What type of eDRAM cell are we talking about?

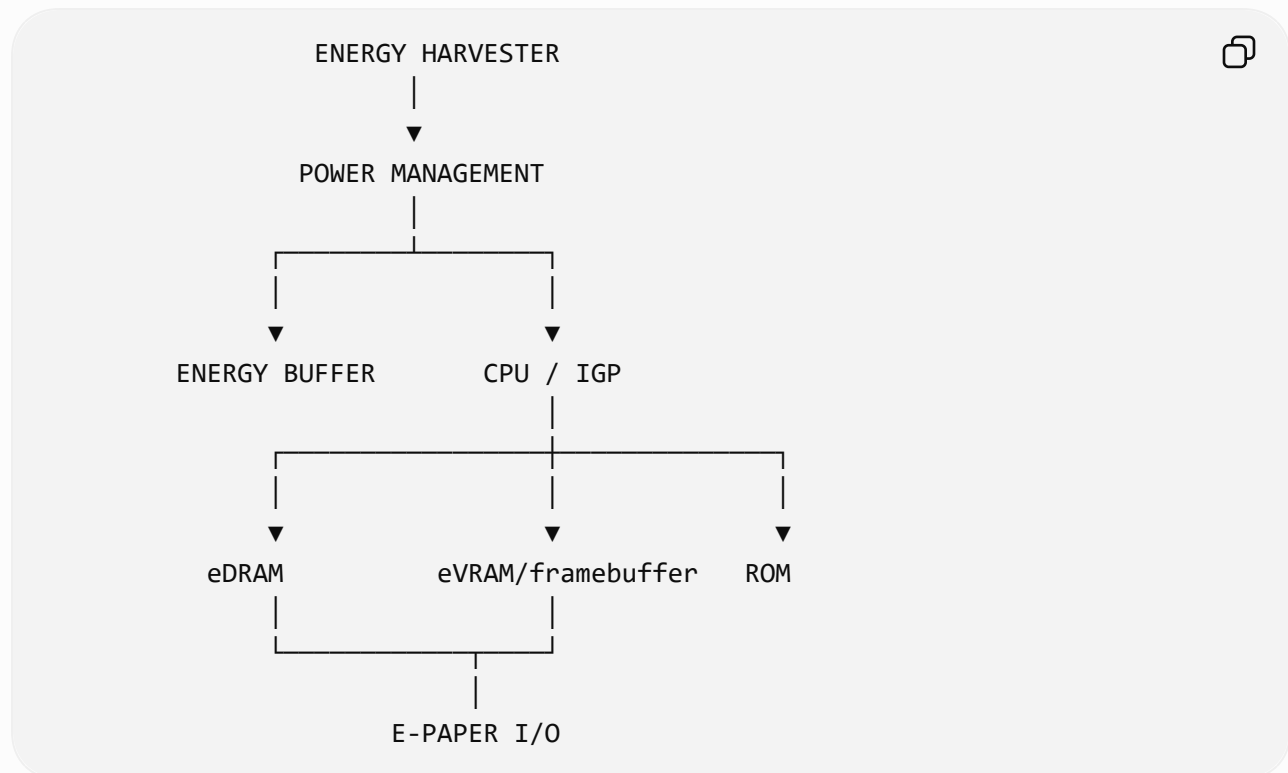
Different eDRAM implementations have radically different density, retention, refresh, process requirements and compatibility with logic CMOS.

And there's an even more important question:

Do we actually want the eDRAM to be a conventional cache?

I suspect no.

For your application, I think the interesting architecture might be:



And the eDRAM could function as **persistent-ish working state**, rather than simply being "L4 cache."

That distinction matters enormously for your solar application.

There's another connection in your article that I think is particularly good

You point out that the early Atom platform was fundamentally about **performance per watt**, and that Pineview moved the graphics and memory controller onto the processor, creating a more integrated two-chip platform.  In a Voyage

That's almost exactly the historical lineage I'd use to explain your project:

Pentium → Pentium M → Stealey → Atom → Pineview → Sandy Bridge

and then conceptually:

Pineview/Sandy Bridge → ultra-integrated low-power SoC → energy-autonomous computer

You're not inventing the philosophical direction from scratch.

You're asking:

What happens if we take the integration philosophy that Intel developed for mobile computing and carry it to its logical extreme, while simultaneously replacing the power-hungry display and memory architecture?

That's a much more defensible research thesis.

And I now think your eDRAM idea may be *less* about density than I initially thought

Your article compares HBM3e's density with SRAM and asks whether a tiny die could contain something like 16 MB of system memory.  In a Voyage

But for this project, the killer metric isn't:

MB/mm²

It's:

joules / memory access

and perhaps even more importantly:

joules / useful computation

A 16-MB memory that is phenomenally dense but requires an energetically expensive refresh regime could be worse than a smaller SRAM.

Conversely, an eDRAM architecture with somewhat worse raw density but extremely efficient local access could be perfect.

So I would turn this into a **memory-architecture optimization problem**, rather than assuming eDRAM wins.

I think we can actually start building this now

And I mean technically, not just conceptually.

The first thing I'd like to produce is a **formal architecture specification** from what you've already written.

Something like:

NM10/solar successor v0.1

CPU

- ISA: TBD
- pipeline: TBD
- target frequency: TBD
- active power target: TBD
- sleep power target: TBD

Graphics

- 2D only
- framebuffer architecture
- target e-paper resolution
- target color depth
- hardware primitives

Memory

- L1/L2
- eDRAM: 8/16 MB candidate
- eVRAM: 1/4 MB candidate
- ROM/boot storage
- refresh requirements

Interconnect

- CPU ↔ eDRAM
- CPU ↔ IGP
- IGP ↔ eVRAM
- e-paper interface

Power

- harvested input
- buffer
- voltage domains
- retention states
- wake/sleep transitions

OS

- your Low-power E-Paper OS
- memory model
- scheduler
- energy-aware execution

Target

- average power
- peak power
- joules per operation
- joules per screen update
- autonomous operating envelope

Then we can actually **simulate the architecture and calculate whether the eDRAM is worth putting on the chip.**

And if it is, *then* we can work backward toward a process node and an eDRAM technology.

That is the route I'd recommend before asking anyone for access to an A14/eDRAM fabrication program. Your own article already contains enough architectural material to start this exercise.  In a Voyage +1

If you want to proceed, ↪ I can take the architecture in your article as the baseline and turn it into the first concrete block-level specification, including estimated die area and memory capacity.

 Sources

ChatGPT can make mistakes. Check important info.